1. $X \sim \text{Exp}(1/25) \Rightarrow f_X(x) = \frac{1}{25} e^{-x/25} \Rightarrow F_X(x) = 1 - e^{-x/25}$

(i) $P(X \le 14) = F(14) = 1 - e^{-14/25} = \text{pexp}(14, 1/25)$

(ii) $F_X(x) = 1 - 0.32 \Rightarrow e^{-x/25} = 0.32 \Rightarrow x = -25 \ln(0.32) = q\text{exp}(0.68, \frac{1}{25})$

(iii) $X \sim \text{Unif}(9, 17)$ is continuous $\Rightarrow P(X = 11) = 0$.

(iv) $P(11.5 < X < 14.5) = \int_{11.5}^{14.5} \frac{1}{8} dx = \frac{3}{8} = \text{punif}(14.5, 9, 17) - \text{punif}(11.5, 9, 17)$

(v) $P(X > 15 \mid X > 14) = \frac{(2)(1/8)}{(3)(1/8)} = \frac{2}{3} = \frac{1 - \text{punif}(15, 9, 17)}{1 - \text{punif}(14, 9, 17)}$

married jurors $X \sim \text{Hyper}(15, 10, 12)$

(vi) $P(X = 6) = \binom{15}{6}\binom{10}{6} / \binom{25}{12} = \text{dhyper}(6, 15, 10, 12)$

(vii) $P(X \le 4) = \left[\binom{15}{2} + \binom{15}{3}\binom{10}{1} + \binom{15}{4}\binom{10}{2}\right] / \binom{25}{12} = \text{phyper}(4, 15, 10, 12)$

(viii) $\text{dgeom}(9, 0.14) = (0.86)^9 (0.14)$ (if you interpret as 10 draws total)

(ix) $\text{pbinom}(3, 10, 0.14) = \sum_{k=0}^{3} \binom{10}{k} (0.14)^k (0.86)^{10-k}$

2. (a) For $x \ge 0$,

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt = \int_0^x e^{-t} dt = -e^{-t} \Big|_0^x = 1 - e^{-x}$$

(b) For $y \ge 0$,

$$F_Y(y) = P(Y \le y) = P(16X^2 \le y) = P\left(X^2 \le \frac{y}{16}\right)$$
$$= P\left(X \le \frac{\sqrt{y}}{4}\right) = F_X\left(\frac{\sqrt{y}}{4}\right) = 1 - e^{-\sqrt{y}/4}$$

(c) For $y > 0$,

$$f_Y(y) = \frac{d}{dy} F_Y(y) = -e^{-\sqrt{y}/4} \cdot \frac{d}{dy}\left(-\frac{1}{4} y^{1/2}\right) = \frac{1}{8\sqrt{y}} e^{-\sqrt{y}/4}$$

3. (a)

value x	1	2	3
$\Pr(X = x)$	0.17	0.44	0.39

gives the marginal dist. $f_X(x)$ at $x = 1, 2, 3$.

(b) $E(X) = (1)(0.17) + (2)(0.44) + (3)(0.39) = 2.22$

(c) $E(XY) = \sum xy\, P(X=x, Y=y)$

$= 2(0.03) + 3(0.07) + 4(0.15 + 0.07) + 6(0.08 + 0.11) + 8(0.21) + 9(0.09) + 12(0.19)$

$= 7.06$

$Cov(X,Y) = E(XY) - E(X)E(Y) = 7.06 - (2.22)(3.18) = 0.0004$

(d) $Cov(XY) \neq 0$ (though it's close), so X, Y are not independent.

If $Cov(XY) = 0$, we are unable to conclude anything about independence.

(e)

value x	1	2	3
$\Pr(X = x)$	0.1034	0.5172	0.3793

gives the conditional dist. $f_{X|Y=2}(x)$ at $x = 1, 2, 3$.

4. At each x, $h(x) = (1-\alpha) f(x) + \alpha g(x) \geq 0$, since $\alpha, f(x), 1-\alpha, g(x) \geq 0$.

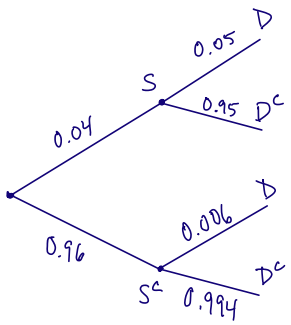
And $\int_{-\infty}^{\infty} h(x)\,dx = \int_{-\infty}^{\infty} \left[(1-\alpha) f(x) + \alpha g(x)\right] dx = (1-\alpha) \int_{-\infty}^{\infty} f(x)\,dx + \alpha \int_{-\infty}^{\infty} g(x)\,dx$

$= (1-\alpha) \cdot 1 + \alpha \cdot 1 = 1.$

5. Policy holder chosen at random.

S = "smoker"
D = "dies in the year"

$P(S) = 0.04$
$P(D \mid S^c) = 0.006$
$P(D \mid S) = 0.05$



$P(S \mid D) = \dfrac{P(D|S)P(S)}{P(D)} = \dfrac{(0.04)(0.05)}{(0.04)(0.05) + (0.96)(0.006)}$

$= 0.2577$

6. $E(Y) = E(3 - 4X) = E(3) - 4E(X) = 3 - 4(11) = -41$

$Var(Y) = Var(3 - 4X) = Var(-4X) = 16\, Var(X) = 48$